FIELD SCAN · CONTINUAL LEARNING

corpus snapshot · 17 OCT 2026 · openalex + arxiv

CONTINUAL LEARNING FROM CHINA.

中国持续学习研究前沿图谱 · 2025–2026

What 1,009 papers say the field is actually doing. Replay is no longer the frontier — the unit of research has become the subspace you may update, the expert that owns a task, and the memory an agent keeps.

THE ONE-LINE SHIFT

From "what to store" to "which weights may change": 150 of 956 papers now allocate capacity in a chosen subspace instead of replaying data.

LoRA/PEFT 150 · replay 148
distillation 119 · regularisation 94
architecture 92 · merging 24

1,009
CS-primary works: "continual learning", ≥1 mainland-China author, Jan 2025 – Oct 2026

531
in 2026 alone (Jan–Oct), vs 425 across all of 2025 — the field is accelerating

30%
of works are LLM-centric — 31.5% in 2026, 27.5% in 2025; vision is still the largest lane

0
median citations — the frontier is pre-consensus, not settled science

01 THREE SHIFTS

METHOD

Replay lost the frontier.

148 papers still replay. But the 2025–26 method wave is capacity allocation: LoRA/PEFT (150), regularisation (94), architecture (92). The question changed from what to store to which subspace may change.

LoRA · null-space · singular vectors

SETTING

CL moved onto pretrained models.

300 vision-flavoured works, mostly frozen encoders plus task-specific projections, adapters or experts. Exemplar-free is the default, and class-incremental toy benchmarks are no longer the main stage.

frozen backbone · adapters · MoE

OBJECT

The object is knowledge itself.

LLM adaptation, agent memory, model merging (24 papers), DP-CL, unlearning. Forgetting is now measured on capabilities — instruction following, reasoning, tool use — not only on labels.

LLM · agents · DP · merging

02 SIX TECHNICAL MOVES

Subspace geometry replaced replay buffers. 01

Null-space projection (O-LoRA lineage) fixes directions, but finite-precision updates still leak into old subspaces. 2026 work turns the step size into a function of accumulated rank — PaLoRA derives $s^* = \sqrt{(R/c)}$ — and moves the allocation unit from layers to singular-vector channels (SVC).

→ PaLoRA · SVC · LoRA-

Orthogonality is not free. 02

Strict isolation blocks transfer between related tasks; CoDe-LoRA names this the orthogonality dilemma. The fix is partial isolation: adaptive null-space projection plus semantic routing, i.e. shared versus private capacity allocated per task, not per parameter.

→ CoDe-LoRA

Isolation by construction: block-diagonal experts. 03

Monarch-MoE replaces dense LoRA experts with block-diagonal factors joined by a fixed permutation: blockwise gradient isolation while keeping 25–50% of the original adapter parameters. CL-MoE routes at task and instance level; SMoLoRA routes at capability level, separating visual understanding from instruction following.

→ Monarch-MoE · CL-MoE · SMoLoRA

Multimodal forgetting is dual. 04

Continual visual instruction tuning degrades visual understanding and instruction following separately (SMoLoRA); CLIP-based CL adds a drifting modality gap (Mind the Gap, ICCV'25). The response is separate routers and explicit gap compensation, not one regulariser.

→ SMoLoRA · Mind the Gap

Agents: memory became the CL substrate. 05

Hippocam consolidates experience along an intent hierarchy; TSRL supervises only failure states and bounds cumulative updates — a rare formal forgetting argument for agent post-training. EmbodiedCLBench supplies the missing evaluation for self-evolving agents.

→ TSRL · Hippocam · EmbodiedCLBench

Generative and private CL are next. 06

Continual diffusion needs gradient projection plus iterative SVD to survive the Markov memory cost; condition-anchored distillation replaces stored prompts with teacher rollouts; DP-SPA shows privatised LoRA training increases forgetting during later adaptation.

→ One-for-More · CAGD · DP-SPA

03 TWELVE PAPERS TO KNOW

LLM · SUBSPACE — arXiv 2610.04226 · Oct 2026

PaLoRA: a derived pacing law

Peking U + CAS

The optimal step restriction under null-space projection is not a tuned constant: it grows with the effective rank R of accumulated updates, $s^* = \sqrt{(R/c)}$, because finite-precision updates leak along multiple directions.

→ Pacing law derived, not tuned; +4 accuracy on 50-task ImageNet-A/R.

LLM · ISOLATION — arXiv 2610.08312 · Oct 2026

CoDe-LoRA and the orthogonality dilemma

BUPT

Strict orthogonal isolation blocks transfer between related tasks. Splits learning into consolidating universal knowledge (adaptive null-space projection) and decoupling task-specific knowledge (semantic routing); replay-free.

→ Best average accuracy across 4 backbones × 3 CL benchmarks.

LLM · CHANNELS — arXiv 2610.11076 · Oct 2026

SVC: singular-vector channels

Beihang + CAS ICT

Each singular-vector channel becomes the unit of stability and plasticity: adaptation benefit is scored against forgetting cost, then channels are chosen by knee screening, Pareto filtering and Otsu thresholding.

→ "Where to adapt" becomes a measured allocation, not a heuristic.

VISION · EXEMPLAR-FREE — CVPR 2025 · doi 10.1109/cvpr52734.2025.01426

LoRA- and the drift-resistant space

Sun Yat-sen U

Subtracts old-task LoRA weights from the pretrained weight to construct a drift-resistant space, so each new task trains against a space that tracks feature drift — with no exemplars and no outdated statistics.

→ Exemplar-free CL with no replay, prototypes or old-task statistics.

MULTIMODAL · CVIT — ICCV 2025 · doi 10.1109/iccv51701.2025.00024

SMoLoRA: dual catastrophic forgetting

Hefei UT + Tsinghua

Continual visual instruction tuning degrades visual understanding and instruction following separately. A separable mixture of LoRA experts with two routers adapts each capability through its own path.

→ Reframes CVIT forgetting as two coupled failures, not one.

MULTIMODAL · EXPERTS — arXiv 2610.13215 · Oct 2026

Monarch-MoE: block-diagonal experts

CAS ICT + UCAS + ShanghaiTech

Replaces dense LoRA experts with block-diagonal factors tied by a fixed permutation: blockwise gradient isolation, all input–output blocks still connected, with batched and fused kernels for memory-limited hardware.

→ Forgetting 0.52 pp / 0.10 pp on first task with 50–75% fewer adapter parameters.

AGENTS · RL — arXiv 2610.17726 · Oct 2026

TSRL: supervise only failure states

CAS Institute of Automation

RL is the optimisation backbone; SFT switches on only at hard states where the policy fails. Restricting supervision to failure states yields an upper bound on cumulative parameter updates, so forgetting is reduced relative to full SFT.

→ Gains on ALFWorld/WebShop/Deep Search at 7B and 32B; +9.6 pts retained capability over SFT→RL.

AGENTS · MEMORY — arXiv 2610.12124 · Oct 2026

Hippocam: intent-structured memory

CAS IIE + UCAS

Structures an agent's work as nested intents: completed intents consolidate to the outcomes needed later, while recursive prefix consolidation makes long-unused experience progressively abstract, with detail recoverable on demand.

→ Lifelong learning as memory organisation, not weight updates.

EMBODIED · BENCHMARK — arXiv 2610.13021 · Oct 2026

EmbodiedCLBench

Peking U + BIGAI

The first embodied-simulation benchmark for continual learning, built on compositional generalisation: agents learn basic skills, then must recombine them to solve unseen advanced tasks.

→ Experience helps consistently — but the gain does not scale with samples.

GENERATIVE · DIFFUSION — CVPR 2025 · doi 10.1109/cvpr52734.2025.00449

One-for-More: continual diffusion

ECNU + Xiamen U + Tencent

Gradient projection protects learned knowledge during continual anomaly detection, while an iterative SVD exploits transitivity of linear representations to cut the memory cost of the diffusion Markov process.

→ Continual generative modelling without per-task replay buffers.

THEORY · NEURO — Nature Communications 2025 · doi 10.1038/s41467-025-56405-9

CH-HNN: corticohippocampal circuits

Tsinghua

Corticohippocampal dual representation in silicon: an ANN stores specific memories and an SNN generalises them, with episode inference for new concepts. Task-agnostic, memory-flat, and cheap at SNN energy cost.

→ A biological circuit used as an architectural prior for CL.

MULTIMODAL · SYNTHETIC — CVPR 2025 · doi 10.1109/cvpr52734.2025.00268

GIFT: synthetic data as replay

Wuhan U + CAS IoA

A diffusion model recreates both pre-training and past-task image–text pairs, and the VLM distils on the synthetic pairs — protecting pre-training knowledge when the original data is gone.

→ Replay without storing anything real: generative buffers replace exemplars.

ALSO ON THE RADAR

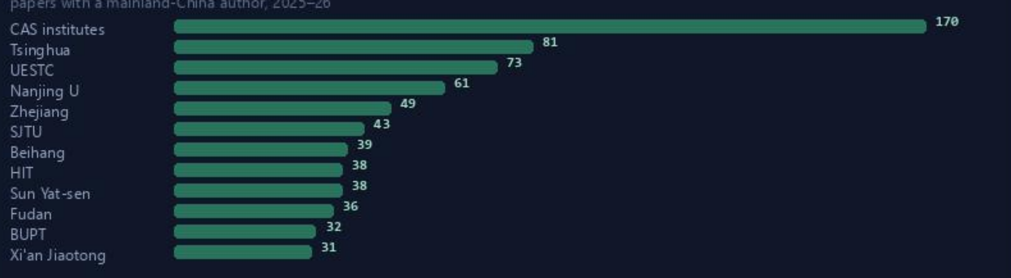
- ChainLoRA: task-vector merging as geometry
  SJTU + HUST · Sep 2026
- GORP: the low-rank constraint is the bottleneck
  ACL 2025
- Knowledge in Superposition: why editing fails
  CAS IoA · AAAI 2025
- Forgetting worsens with scale, 1B–14B
  Westlake + Tencent · TASLP 2025
- Beyond Forgetting: continual RLVR
  PKU + BIGAI · Aug 2026
- CMT: compress new knowledge into memory
  HIT · AAAI 2025
- GainLoRA: gated LoRA integration for CL
  Nanjing U · NeurIPS 2025
- NTK theory of PEFT continual learning
  Tianjin + Zhejiang · TPAMI 2026
- A buffer does not inherently reduce forgetting
  HUST/SYSU/NUDT · Neural Networks 2025
- Cumulative-error bound, noisy drifting streams
  SJTU + Nanjing · TPAMI 2025
- EFRE: prompt repertoires beat one prompt
  CUHK-SZ + USTC + Shanghai AI Lab
- DP-SPA: differentially private continual LoRA
  Tianjin + Xidian · Oct 2026
- Mind the Gap: CLIP modality gap in CL
  Nankai · ICCV 2025
- Proof: projection fusion for VLMs
  Nanjing U · TPAMI 2025 · 56 cites
- CL-MoE: dual momentum experts, VQA
  ECNU · CVPR 2025
- RIFAR: forgetting-aware replay for robots
  Tsinghua + Zhejiang · Oct 2026
- CL-VISTA: video-LLM continual benchmark
  CAS IoA · 2026
- Model merging in LLMs / MLLMs: survey
  Sun Yat-sen · ACM CSUR 2026

04 THE CORPUS IN NUMBERS

WHO PUBLISHES

Top 12 institutions

papers with a mainland-China author, 2025–26

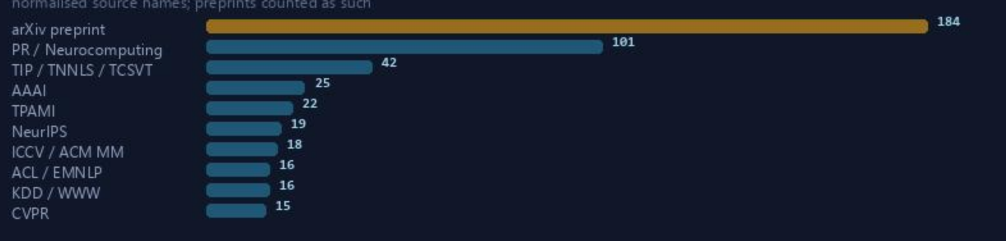


CAS = its institutes rolled up · 491 works name at least one of these

WHERE IT LANDS

Venue mix

normalised source names; preprints counted as such

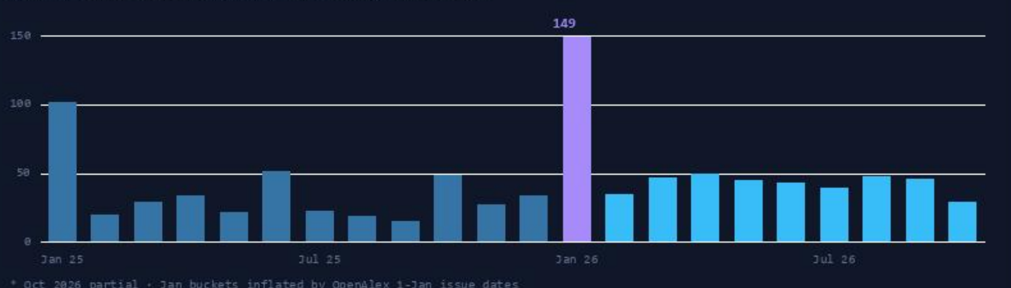


arXiv 184 · 29 surveys · 651 works still uncited

MOMENTUM

Output per month

2026 YTD already beats all of 2025: 531 vs 425, with 2.5 months fewer



* Oct 2026 partial · Jan buckets inflated by OpenAlex 1-Jan issue dates

DIVISION OF LABOUR

Who owns which lane

read from the authorship of the 2026 preprints

CAS IoA	editing theory · agent RL post-training
CAS ICT	structured experts (Monarch-MoE)
Tsinghua · PKU	LoRA-CL core · pacing · surveys
Nanjing U	VLM projection fusion (Proof)
SJTU	task-vector merging geometry
ECNU	multimodal MoE (CL-MoE)
Sun Yat-sen	exemplar-free CL · merging surveys
HIT · Westlake/TC	frozen memory · forgetting baselines
PKU + BIGAI	agent benchmarks

authorship-based, indicative — not an exhaustive census

05 WHAT IS NOT SOLVED

01 No predictive theory of transfer.

Bounds exist per method — pacing laws, update-norm bounds — but none predicts when shared structure helps and when it interferes. Theory is only ~6% of the corpus; "orthogonality dilemma" is a name, not yet a theory.

02 Evaluation is fragmented.

UCIT, MLLM-DCL, EmbodiedCLBench and ALFWorld each measure something different; few report backward transfer, forward transfer and compute together. 651 of 956 works are uncited — no consensus metric.

03 Forgetting of capabilities is unmeasured.

Instruction following, reasoning and tool use degrade without a clean metric; most benchmarks still score task accuracy on labels.

04 Replay still wins on accuracy.

Replay-free methods trail by roughly 10–20 points on class-incremental vision; the memory, privacy and latency costs of replay remain unpriced in most papers.

05 Continual RL and embodiment.

Non-stationary dynamics, sim-to-real transfer and long-horizon credit assignment are unsolved; the benchmarks are new and small.

06 Privacy, unlearning and CL collide.

DP noise compounds across tasks and interacts with LoRA's two factors; the boundary between machine unlearning and continual learning is undefined in both method and regulation.

METHOD & CAVEATS

OpenAlex works whose title or abstract contains "continual learning", primary field Computer Science, with at least one mainland-China institution among the authors, published 2025-01-01 to 2026-10-17. 1,009 records matched (1,333 phrase matches across all fields); 956 unique titles analysed. Counts are multi-label — one paper can be vision and LLM — so shares overlap and do not sum to 100%. Citation counts are OpenAlex at snapshot date; the median is 0, so citation rank is not a quality rank. "Chinese" = any mainland-China institution on the author list, not necessarily the first author. Venue names are normalised from raw source strings; arXiv share reflects the pre-publication frontier.

READING PATH

1. Read the surveys first: TPAMI 2024 CL survey (1,087 cites) as the frame, then multimodal CL (TNNLS 2026), lifelong LLMs (ACM CSUR 2025), CL systematic review (Neural Networks 2025).
2. Pick a benchmark that measures capabilities, not labels: UCIT, MLLM-DCL, CL-VISTA or EmbodiedCLBench.
3. Pick a mechanism. Subspace geometry has the most theoretical traction, block-diagonal experts the most hardware reality, agent memory the least explored.

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